### Photosensitivity

Substances being analyzed in blood can be broken down by light, resulting in false values. One test that is commonly impacted by light is bilirubin. A specimen for bilirubin loses 50% of the bilirubin after 1 hr of light exposure. One way to protect the blood and the substance within the blood is to wrap the tube in aluminum foil. Amber tubes, biohazard bags, and light-blocking transport containers can also transport blood to the laboratory.

## *Clinical Research*

As a phlebotomist involved in clinical research, adhering to laboratory protocols is paramount to ensure the validity and integrity of the study results. Following these protocols meticulously is essential for maintaining the consistency and standardization required in a research setting.

Tube processing is the first critical step. Phlebotomists must be well-versed in the specific requirements for each type of specimen, including the correct order of draw, the type of anticoagulant or preservative in the tube, and the required volume of blood. Proper mixing and handling of the tubes post-collection are also crucial to prevent hemolysis or clotting, which could compromise the sample.

Obtaining the correct specimen bag is another important protocol. Specimen bags often have separate compartments for the specimen tubes and the associated paperwork, which must be filled out accurately and completely. This paperwork typically includes patient identification, time and date of collection, and specific study identifiers.

For samples that must be shipped, an air waybill is required. This document serves as a shipping label and a tracking form, which includes details about the sender, the recipient, and any special shipping instructions. Phlebotomists must ensure that the air waybill is filled out correctly to avoid any delays or mix-ups in transport.

Finally, packaging the specimens for transport must be done in compliance with all safety and regulatory guidelines. This often involves using biohazard bags, absorbent materials, and sturdy shipping containers that are designed to maintain the required temperature and protect the samples during transit. Phlebotomists must be familiar with the regulations for shipping biological materials, including international guidelines if the samples are being sent to a different country.